

Practice Midterm Exam

CAS CS 132: Geometric Algorithms

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- ▷ You will have approximately 75 minutes to complete this exam. Make sure to read every question, some are easier than others.
- ▷ Do not remove any pages from the exam.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.
- ▷ We will not look at any work on the pages marked "*This page is intentionally left blank.*" You should use these pages for scratch work.

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1 Linear Equations and Spans

(6 points) Determine a linear equation whose solution set is the span of the following two vectors. (*Note.* You may not use the cross product for this problem.)

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ -2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & b_1 \\ -1 & -2 & b_2 + b_1 \\ 2 & 3 & b_3 - 2b_1 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & b_1 \\ 0 & 1 & b_2 + b_1 + 3(b_1 + b_2) \\ 0 & -3 & b_3 - 2b_1 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & b_1 \\ 0 & 1 & b_2 + b_1 \\ 0 & 0 & b_3 - 2b_1 + 3(b_2 + b_1) \end{bmatrix}$$

$$b_3 - 2b_1 + 3(b_2 + b_1) =$$

$$b_3 - \cancel{2b_1} + 3b_2 + \cancel{3b_1} =$$

$$b_1 + 3b_2 + b_3 = 0$$

$$1 + 3(-1) + 2 = 1 - 3 + 2 = 0$$

$$3 + 3(-2) + 3 = 3 - 6 + 3 = 0$$

Solution.

$$x + 3y + z = 0$$

2 Pseudoinverses

(6 points) Determine a matrix A and two values k and h such that

$$A \begin{bmatrix} 1 & -1 \\ 3 & -2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & -1 \\ 3 & -2 \\ -1 & 3 \end{bmatrix} A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ h & k & 0 \end{bmatrix}$$

(Hint. First determine the rightmost column of A .)

$$\begin{bmatrix} 1 & -1 \\ 3 & -2 \\ -1 & 3 \end{bmatrix} A = \begin{bmatrix} 1 & -1 \\ 3 & -2 \\ -1 & 3 \end{bmatrix} [\vec{a}_1 \ \vec{a}_2 \ \vec{a}_3] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ h & k & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ 3 & -2 \\ -1 & 3 \end{bmatrix} \vec{a}_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 1 & -1 & 0 \\ 3 & -2 & 0 \\ -1 & 3 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \vec{a}_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ 3 & -2 \\ -1 & 3 \end{bmatrix} \vec{a}_1 = \begin{bmatrix} 1 \\ 0 \\ h \end{bmatrix} \quad \begin{bmatrix} 1 & -1 & 1 \\ 3 & -2 & 0 \\ -1 & 3 & h \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -3 \\ 0 & 2 & h+1 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & h+7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -3 \\ 0 & 0 & h+7 \end{bmatrix} \quad \vec{a}_1 = \begin{bmatrix} -2 \\ -3 \\ h \end{bmatrix} \quad h = -7$$

$$\begin{bmatrix} 1 & -1 \\ 3 & -2 \\ -1 & 3 \end{bmatrix} \vec{a}_2 = \begin{bmatrix} 0 \\ 1 \\ k \end{bmatrix} \quad \begin{bmatrix} 1 & -1 & 0 \\ 3 & -2 & 1 \\ -1 & 3 & k \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & k \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & k-2 \end{bmatrix} \sim$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & k-2 \end{bmatrix} \quad \vec{a}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad k = 2$$

Solution.

$$A = \begin{bmatrix} -2 & 1 & 0 \\ -3 & 1 & 0 \end{bmatrix}$$

$$h = -7$$

$$k = 2$$

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3 Linear Transformations

Consider the following linear transformations.

$$S\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2x_1 \\ -x_2 \\ x_1 + x_2 \end{bmatrix} \quad T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_2 + x_3 \\ -x_1 - 2x_2 - 3x_3 \end{bmatrix}$$

- A. (3 points) Determine the matrix that implements $S \circ T$, the composition of S and T . That is, determine the matrix A such that the transformation $\mathbf{x} \mapsto S(T(\mathbf{x}))$ is equivalent to the transformation $\mathbf{x} \mapsto A\mathbf{x}$.

$$S\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \quad S\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \quad S(\vec{x}) = \begin{bmatrix} 2 & 0 \\ 0 & -1 \\ 1 & 1 \end{bmatrix} \vec{x}$$

$$T\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad T\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \quad T\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \end{bmatrix} \quad T(\vec{x}) = \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \end{bmatrix} \vec{x}$$

$$\begin{bmatrix} 2 & 0 \\ 0 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -1 & -2 & -3 \end{bmatrix} =$$

$$(S \circ T)(\vec{x}) = \begin{bmatrix} 2 & 2 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -2 \end{bmatrix} \vec{x}$$

Solution.

$$\begin{bmatrix} 2 & 2 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -2 \end{bmatrix}$$

- B. (6 points) Determine a general form solution that describes the preimages of the following vector $\mathbf{b}$ under the linear transformation $S \circ T$. That is determine a general form solution where $S(T(\mathbf{x})) = \mathbf{b}$ if and only if $\mathbf{x}$ is in the solution set described by your general form solution.

$$\mathbf{b} = \begin{bmatrix} 4 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{aligned} \begin{bmatrix} 2 & 2 & 2 & 4 \\ 1 & 2 & 3 & 1 \\ 0 & -1 & -2 & 1 \end{bmatrix} &\sim \begin{bmatrix} 1 & 1 & 1 & 2 \\ 1 & 2 & 3 & 1 \\ 0 & -1 & -2 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & -1 & -2 & 1 \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 & 3 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Solution.

$$x_1 = 3 + x_3$$

$$x_2 = -1 - 2x_3$$

$$x_3 \text{ is free}$$

4 True/False

Determine if each of the following statements is **True** or **False**. Bubble in your answers below. You do not need to show your work.

- A. (1 point) A system of linear equations with 3 variables and 4 equations cannot be consistent.
- ☐ True
☒ False
- B. (1 point) For any matrix $A \in \mathbb{R}^{n \times n}$, if $A = LU$ where L and U form an LU-factorization of A , and if L is invertible, then so is A .
- ☐ True
☒ False
- C. (1 point) For any matrices $A, B \in \mathbb{R}^{n \times n}$, if A and B are invertible, then so is AB .
- ☒ True
☐ False
- D. (1 point) If A is the augmented matrix of a linear system and it has a pivot position in every column, then the system is inconsistent.
- ☒ True
☐ False
- E. (1 point) If the equation $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution, then it has infinitely many solutions.
- ☒ True
☐ False
- F. (1 point) For any vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbb{R}^n$, if $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent, then so is $\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_3\}$.
- ☐ True
☒ False
- G. (1 point) For any matrix $A \in \mathbb{R}^{m \times n}$, the matrix AA^T is symmetric (recall that a matrix B is symmetric if $B = B^T$).
- ☒ True
☐ False
- H. (1 point) For any matrix $A \in \mathbb{R}^{m \times n}$, if $m < n$, and the columns of A span $\mathbb{R}^m$, then the reduced echelon form of A has only 0s and 1s.
- ☐ True
☒ False
- I. (1 point) For any matrices $A, B \in \mathbb{R}^{n \times n}$, if $AB = I$, then $BA = I$.
- ☒ True
☐ False

5 Linear Dependence

(6 points) Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ be vectors such that

$$[\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3 \ \mathbf{v}_4] \sim \begin{bmatrix} 2 & -3 & 3 & 8 \\ 0 & -2 & -2 & 7 \\ 0 & 0 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Determine a dependence relation for the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$. That is, determine coefficients $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ such that $\alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \alpha_3 \mathbf{v}_3 + \alpha_4 \mathbf{v}_4 = \mathbf{0}$.

$$\begin{aligned} &\sim \begin{bmatrix} 2 & -3 & 3 & 0 \\ 0 & -2 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & -3 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & 0 & 6 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & 0 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{aligned} x_1 &= -3x_3 \\ x_2 &= -x_3 \\ x_3 &\text{ is free} \\ x_4 &= 0 \end{aligned} \end{aligned}$$

Solution.

$$3\vec{v}_1 + \vec{v}_2 - \vec{v}_3 = \vec{0}$$

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